

Project Overview

HomeHub is a sophisticated, multi-role Android application designed to streamline property management, caretaker services, and student living. It integrates a robust Firebase backend with a modular Kotlin-based frontend to provide a seamless experience for all stakeholders in the residential ecosystem.

Technical Architecture

- **Platform:** Android (Kotlin)
- **Backend:** Firebase (Authentication, Real-time Database, Storage)
- **Architecture:** Modular Package-by-Feature
- **Automation:** Python-based maintenance scripts

User Roles & Modules

Student / Resident

Property search, application, rent payment, and maintenance request tracking.

Caretaker

Maintenance task management, resident communication, and property inspection logs.

Supplier

Service request fulfillment, invoicing, and payment tracking.

Administrator

User role management, system monitoring, and financial oversight.

Key Features

- **Real-time Chat:** Instant communication between all user roles.
- **Billing System:** Integrated rent and service tracking.
- **Maintenance Workflow:** End-to-end service request management.
- **Firebase Integration:** Secure sync and authentication.

Setup Instructions

1. Clone the repository and ensure all assets are present.
2. Place `serviceAccountKey.json` in the root directory.
3. Run `RUN_ME_FIRST.bat` to initialize the environment.
4. Open in Android Studio and sync Gradle.